Introduction to Agile Scrum

Software Development Practice

- Lean
- Agile Scrum

Software Development Life Cycle/Project Methodologies

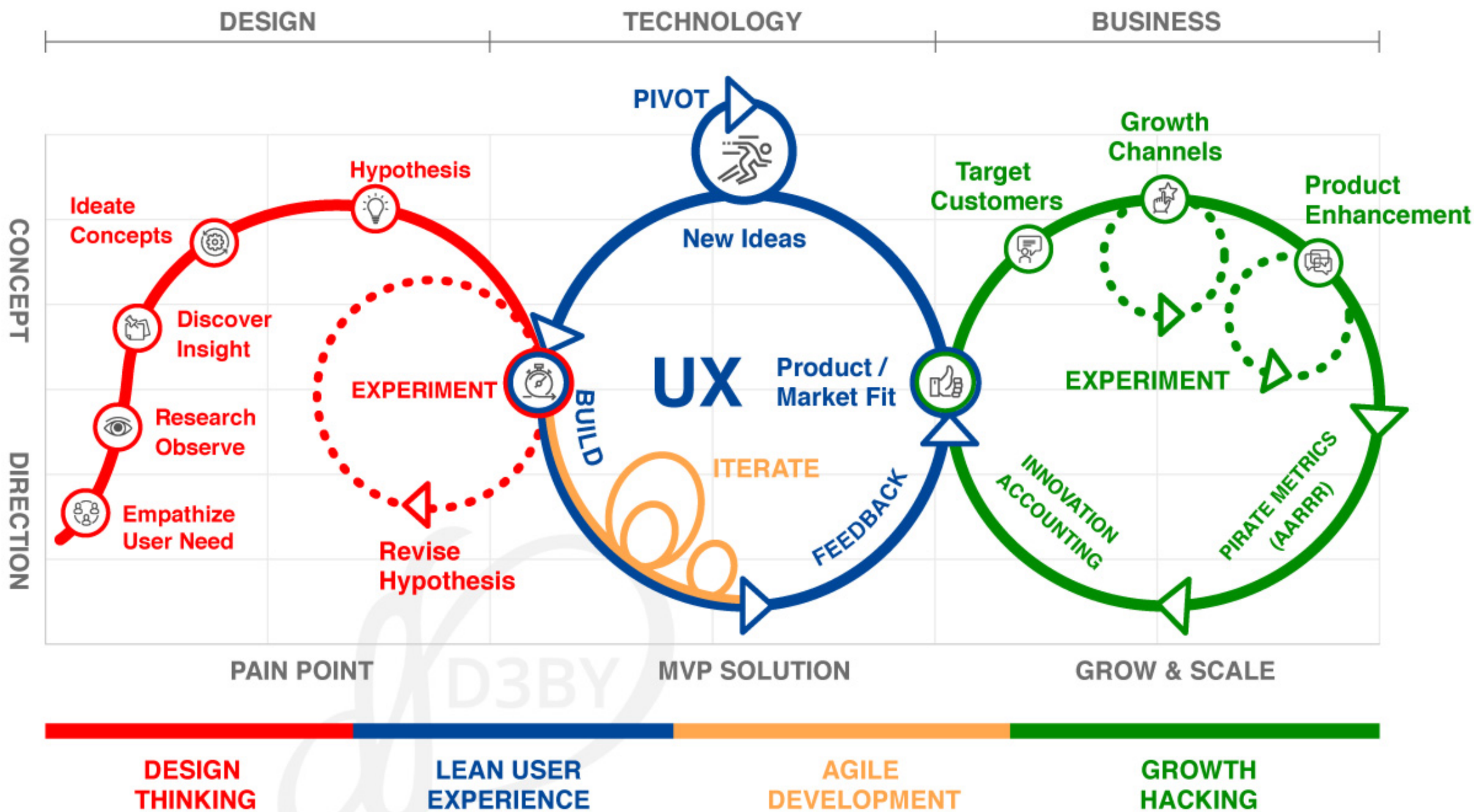
- Traditional
  - Waterfall
  - Incremental / Waterfall (modified)
  - Prototyping
  - Rapid Application Development (RAD)
  - Spiral
- Agile
  - **Scrum**, **Kanban**, eXtreme Programming (XP), Lean Development, Feature Driven Development (FDD), DevOps, Test Driven Development (TDD)

Agile & Lean

Software Development Practice

Software Philosophies & Methodologies

Philosophies	Lean, Agile, Traditional etc
Methodologies	Scrum, Kanban, XP, TPS etc…
Tools	sprints, boards, tests, cohort analysis


DESIGN
TECHNOLOGY
BUSINESS
PIVOT
New Ideas
Hypothesis
Ideate Concepts
Discover Insight
Research Observe
Empathize User Need
EXPERIMENT
Revise Hypothesis
BUILD
UX
Product / Market Fit
ITERATE
FEEDBACK
Target Customers
Growth Channels
Product Enhancement
EXPERIMENT
INNOVATION ACCOUNTING
PIRATE METRICS (AARRR)
CONCEPT
DIRECTION
PAIN POINT
MVP SOLUTION
GROW & SCALE
DESIGN THINKING
LEAN USER EXPERIENCE
AGILE DEVELOPMENT
GROWTH HACKING

Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and Interactions	over	processes and tools	People
Working software	over	comprehensive documentation	Results
Customer collaboration	over	contract negotiation	Collaboration
Responding to change	over	following a plan	Responsiveness

http://www.agilemanifesto.org/

12 Principles for Agile

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.	Working software is the primary measure of progress.
Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.	Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.	Continuous attention to technical excellence and good design enhances agility.
Business people and developers must work together daily throughout the project.	Simplicity--the art of maximising the amount of work not done--is essential.
Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.	The best architectures, requirements, and designs emerge from self-organising teams.
The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.	At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behaviour accordingly.

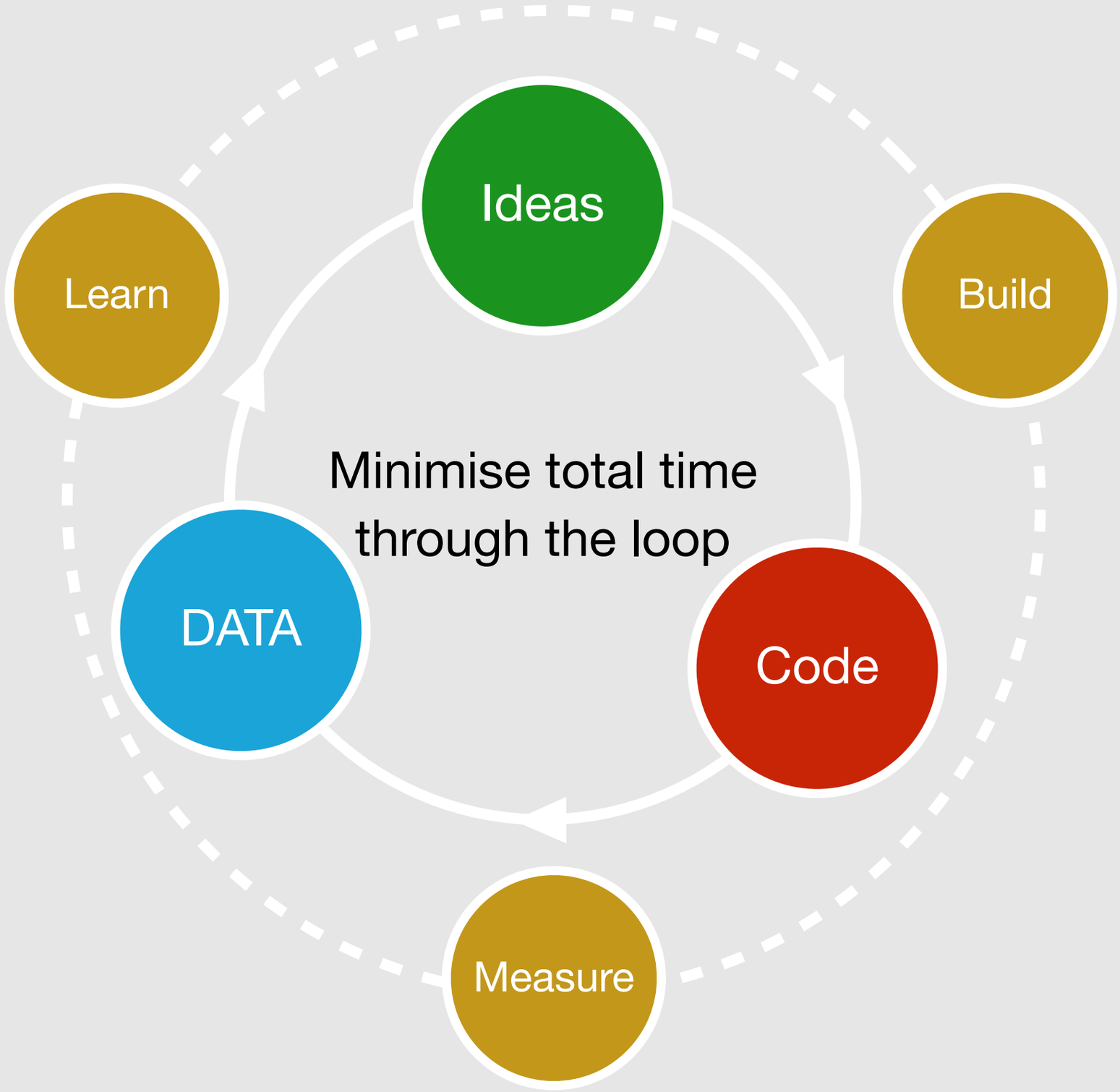
Lean Principles

Eliminate Waste	Partially done work Extra processes Extra features Task switching	Waiting Motion Defects Management activities
Amplify learning	Short iteration cycles for continuous learning Feedback from customers	
Decide as late as possible	Decisions based more on facts Avoid uncertainties and predictions	
Deliver as fast as possible	Short interactions Faster feedback Ability to delay decisions	
Empower the team	Choose their own commitments Developers involved in decision making Given access to the customer	
Build integrity in	Refractor code Reuse code Automated building and testing	
See the whole	Systems are not just a sum of their parts. Think big, act small, fail fast, learn rapidly	

What is Lean?

- Maximise Customer Value
  - Value Proposition Canvas
- Eliminate waste/resources
  - Time
  - Process
  - Effort
  - Capital
  - Resources
  - Intellect
- Flow of User Value
- Perfection - repeat until truly Lean

Lean


Ideas
Learn
Build
Minimise total time
through the loop
DATA
Code
Measure

Lean vs Agile

Agile Philosophy	Lean Philosophy
Is aimed at executing tasks faster, adapting to changes easier	Is about Smart development, when you improve virtually everything you do by eliminating anything that doesn't bring value to the customer
Makes the developing process flexible	Makes the developing process sustainable
Was initially designed for Software Development, then expanded to Marketing, and is currently applied in other areas	Started from traditional manufacturing and expanded to all existing industries
Action loop: product backlog – sprint backlog – iteration (sprints) – potentially shippable result	Action loop: build-measure-learn
Method for demonstrating progress — definition of 'done'	Method for demonstrating progress — validated learning
Toolkit: sprints, boards, Scrum Master, acceptance tests, user story mapping etc.	Toolkit: hypotheses, split (A/B) tests, customer interviews, funnel and cohort analysis, Customer Success Manager etc.